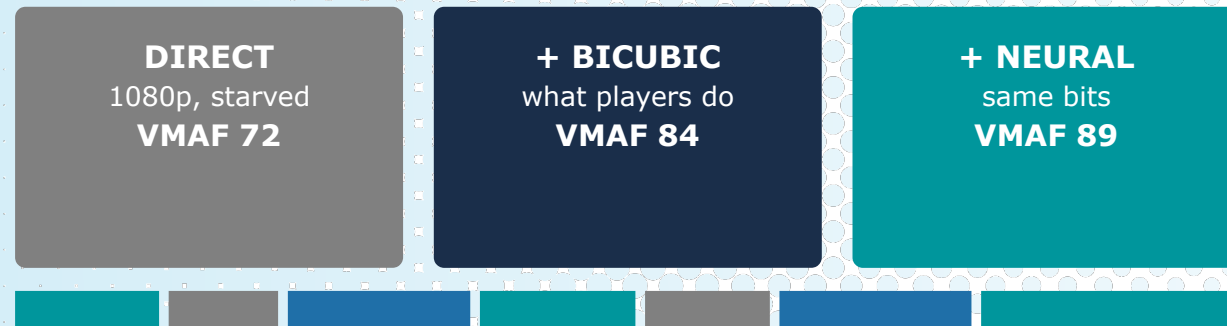


Send fewer bits. Let the player put them back.

Build a neural-enhanced video delivery pipeline · Ofinno · CS491 sponsored project

THE DEMO YOU WILL BUILD



Above: the chain your compiler picked per segment. Numbers illustrative.

WHAT YOU BUILD

PACKAGER encode one compact base, measure every segment, write a manifest

COMPILER pick base and enhancement chain per segment, per device budget

HARNESS three anchors, VMAF and VMAF-NEG, one command to rebuild the report

Python-first, laptop-only, offline. No codec to write, no model to train: pretrained models used as black boxes.

HOW SUCCESS IS DEFINED

TIER 1 · it runs end to end
base encode, enhancement, two output profiles, first honest curves (Fall PoC)

TIER 2 · the MVP
per-segment measured cost table, content-adaptive compiler, comparison interface, reproducible report

TIER 3 · your call
real-time inference, short-form output, per-genre frontier study, HLS/DASH packaging

The question you get to answer

How much does neural enhancement buy over plain bicubic upscaling, on which content, at what compute cost?

WHAT OFINNO SUPPLIES

- A genre-diverse, license-checked clip set
- A curated allowlist of pretrained models
- Cost-table template and harness guidance
- Biweekly meetings with our engineers